PALADINMEDIA — SPRINT & SCRUM PROJECT LOG

Project: Post-Quantum Media Steganography System (MVP 0 & Phase 2)
Member / Owner: Rohit Gautam | Status: Completed & Verified

1. Sprint-1 Backlog: Cryptographic Foundation & Payload

User Story	Task Description	Owner	Status
Setup & Core Architecture	Initialize repository layout, core exception hierarchy (src/core/exceptions.py), and configure pyproject.toml with cryptography and reedsolo dependencies.	Rohit Gautam	Done
Payload Serialization & Framing	Implement SecretSerializer (src/core/serialization/serializer.py) with canonical binary framing (Magic b'PLDS', version 0x01, media type enum, sorted JSON metadata, CRC-32).	Rohit Gautam	Done
Cryptographic Protection	Develop AESGCMProvider (src/security/crypto/provider.py) implementing AES-256-GCM with fresh 96-bit nonces, 128-bit GHASH tag, AAD context binding, and fail-closed terminal rejection.	Rohit Gautam	Done
Error Correction Coding (ECC)	Implement ReedSolomonEncoder (src/security/ecc/encoder.py) over GF(2^8) with systematic chunking, 2t=32 parity symbols/block, length prefix framing, and symbol-level repair.	Rohit Gautam	Done
Security Pipeline Integration	Create SecurityPipeline (src/core/pipeline.py) enforcing unidirectional sequence: Serialize -> AES-GCM -> Reed-Solomon, and build standalone 24-check verification script (scripts/verify_sprint1.py).	Rohit Gautam	Done
Verification & Test Suite	Implement 4-tier test suite covering T-001 through T-004 in tests/unit and tests/integration (83 tests proving zero plaintext leakage on tampering and boundary handling).	Rohit Gautam	Done

2. Sprint-2 Backlog: Signal Processing, Carrier Infrastructure & UI Observability

User Story	Task Description	Owner	Status
Wavelet Transform Engine	Implement one-level 2D Haar DWT and IDWT (src/carrier/dwt.py) for 512x512 RGB carrier images, outputting explicit (3, 256, 256) subbands with zero reconstruction error.	Rohit Gautam	Done
Capacity Hard Gate & Header	Implement CapacityCalculator (src/carrier/capacity.py) enforcing 8,176 B ceiling, and CarrierHeader (src/carrier/header.py) with 16-byte fixed canonical framing (b'PLDH').	Rohit Gautam	Done
Deterministic Position Mask	Develop DeterministicMaskGenerator (src/carrier/mask.py) using fixed header anchors and non-circular PRNG position sampling derived from session context and payload length.	Rohit Gautam	Done
Transform Embedding & Recovery	Implement BaselineEmbedder and Extractor (src/carrier/) using integer-parity modulation on 2*HH in Red channel, guaranteeing 0.000000 BER through 8-bit PNG reload.	Rohit Gautam	Done
Media-to-Media Pipeline	Build MediaStegoPipeline (src/core/media_pipeline.py) orchestrating complete secret-to-stego-to-recovered workflow with	Rohit Gautam	Done

	PSNR (>75 dB), SSIM (1.0), and exact byte equality.		
Observability API & Streaming	Build FastAPI server (src/api/server.py) and MediaEventAdapter (src/api/event_adapter.py) streaming real-time Server-Sent Events (SSE) and base64 DWT/stego visual artifacts.	Rohit Gautam	Done
Frontend Laboratory Interface	Create React + TypeScript + Vite UI (frontend/) featuring 4-quadrant DWT coefficient inspector, live capacity bar, side-by-side media cards, and telemetry log.	Rohit Gautam	Done
Repo Organization & Testing	Organize repository into single-source-of-truth structure (docs/, src/, frontend/, tests/, scripts/, README.md) with 99 passing tests and E03 experiment record.	Rohit Gautam	Done

3. Weekly Scrum Log

Date	Member	Done since last scrum	Plan for next	Blockers
20 Sep 2026	Rohit Gautam	Completed preflight analysis of governing specifications (Architecture, Product Vision, Systems & Research). Validated MVP 0 Carrier Dataset (60 images) and generated mvp0_manifest.json.	Establish package scaffolding, exception hierarchy, and core serialization wire framing.	Discrepancies between theoretical transform capacity and usable payload budget (Resolved: isolated protocol overheads).
23 Sep 2026	Rohit Gautam	Implemented SecretSerializer (b'PLDS', CRC-32), AESGCMProvider (fail-closed, 0 plaintext leakage), and ReedSolomonEncoder (GF(2^8), t=16). Verified 83 unit/integration tests and passed all 24 checks in verify_sprint1.py.	Transition to Sprint 2 / Phase 2: Design 1-level 2D Haar DWT engine, carrier header, and capacity calculator.	None.
26 Sep 2026	Rohit Gautam	Built HaarDWT2D, 16-byte CarrierHeader, non-circular DeterministicMaskGenerator, and CapacityCalculator hard gate (PAYLOAD_CAPACITY_EXCEEDED). Created authentic 24x24 small secret image fixture (secret_small.png).	Implement transform-domain parity embedding, extraction, and end-to-end media roundtrip testing.	Large 791 KB image payloads failing capacity check (Resolved: enforced hard gate to prevent overflow and selected small image fixture).
29 Sep 2026	Rohit Gautam	Implemented BaselineEmbedder, Extractor, and MediaStegoPipeline. Verified 0.000000 BER through 8-bit PNG reload, PSNR > 75 dB, and 100% exact byte equality. Added tests T-011 through T-020 (99/99 pytest tests passing). Documented E03_media_steganography.md.	Build the visual observability layer: FastAPI SSE stream adapter and React/Vite laboratory UI.	None.
3 Oct 2026	Rohit Gautam	Developed FastAPI streaming server (src/api/server.py) and React + TypeScript + Tailwind frontend (frontend/). Implemented real 4-quadrant DWT view (LL, LH, HL, HH), live capacity meter, and	Complete full repository cleanup, audit .agents/, remove redundant files, and finalize documentation.	Browser Hot Module Reload type import issues during interface initialization (Resolved:

		execution telemetry log.		separated TypeScript type imports).
6 Oct 2026	Rohit Gautam	Completed repository reorganization: moved specs to docs/specifications/, test specs to docs/testing/, frontend to frontend/, scripts to scripts/. Removed stale .agents/ and .docx.txt artifacts. Created root README.md and .gitignore. Verified all 99 tests pass and frontend builds cleanly.	Finalize project deliverables, review experiment E03 metrics, and prepare submission report.	None.